

KVAMSOY, Norway

WMO: 013290

Lat: 60.364N

Lon: 6.279E

Elev: 50

StdP: 100.73

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-4.0	-12.9	1.2	-3.3	-11.3	1.4	-2.2	19.5	3.1	17.9	2.1	6.3	50	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	25.2	17.6	23.3	16.5	21.5	15.7	18.5	23.9	17.4	21.9	16.5	20.4	2.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.7	20.7	15.4	11.0	19.6	14.6	10.5	18.6	52.3	24.0	49.0	21.9	46.4	20.4	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	13.2	11.5	DB	-6.0	28.0	1.9	2.0	-7.3	29.5	-8.4	30.7	-9.5	31.8	-10.8	33.3	(4)
(5)			WB	-7.4	19.7	2.0	1.4	-8.8	20.7	-10.0	21.5	-11.1	22.2	-12.6	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	1.6	1.7	3.6	7.3	11.4	14.6	17.0	15.8	12.7	8.6	5.1	2.6		
	DBStd	6.15	2.56	2.76	2.73	2.87	3.60	2.92	3.23	2.50	2.63	2.55	3.04	3.04		
	HDD10.0	1247	260	232	199	89	24	2	0	0	6	57	149	228		
	HDD18.3	3611	518	466	458	329	216	118	66	86	168	302	398	487		
	CDD10.0	719	0	0	0	10	69	139	217	179	88	14	2	0		
	CDD18.3	40	0	0	0	0	2	6	24	7	1	0	0	0		
	CDH23.3	168	0	0	0	0	13	26	116	14	1	0	0	0		
	CDH26.7	25	0	0	0	0	2	2	21	1	0	0	0	0		
(14) Wind	WSAvg	4.3	5.7	4.6	4.9	3.8	3.6	3.5	3.3	3.2	3.8	4.6	5.2	5.2		
(15) Precipitation	PrecAvg	2257	262	218	216	113	99	119	129	147	208	246	228	270		
	PrecMax	3253	498	476	699	225	215	241	200	249	363	528	473	479		
	PrecMin	1415	3	1	28	32	16	39	40	63	60	69	43	35		
	PrecStd	385	140	133	135	47	52	51	38	51	93	113	116	140		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.6	11.9	17.5	25.2	26.0	29.1	25.3	20.8	16.9	13.0	9.8		
		MCWB	5.6	5.4	6.5	10.5	16.6	16.9	19.1	17.7	14.9	12.3	9.3	7.2		
	2%	DB	7.0	7.4	9.6	15.4	21.5	23.7	26.4	22.7	19.1	14.6	11.3	8.4		
		MCWB	5.3	5.0	5.6	9.4	14.1	15.4	18.7	16.6	14.2	10.8	8.3	6.4		
	5%	DB	5.9	6.5	8.2	13.6	19.3	21.7	24.2	21.1	17.6	13.1	10.5	7.5		
		MCWB	4.0	4.3	5.1	8.4	12.7	14.6	17.6	15.9	13.5	10.3	8.0	5.7		
	10%	DB	5.0	5.5	7.1	12.0	17.1	19.7	22.1	19.7	16.3	11.9	9.4	6.6		
		MCWB	3.3	3.5	4.8	7.4	11.5	13.9	16.9	15.2	12.9	9.3	7.5	4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	6.9	7.5	11.2	16.7	17.7	20.5	18.7	16.0	13.1	10.3	8.5		
		MCDB	7.4	7.8	9.2	16.6	23.9	24.4	27.1	23.7	19.0	14.8	11.3	9.0		
	2%	WB	5.8	5.9	6.6	9.9	14.7	16.2	19.2	17.4	15.0	12.0	9.5	7.2		
		MCDB	6.6	6.9	8.4	13.9	20.8	21.9	25.1	21.4	17.7	14.0	10.8	8.0		
	5%	WB	4.6	4.9	5.9	9.0	13.2	15.2	18.2	16.6	14.3	11.0	8.6	6.1		
		MCDB	5.6	6.1	7.6	12.9	18.4	20.5	23.1	19.9	16.8	12.5	10.0	7.1		
	10%	WB	3.7	3.9	5.1	8.0	12.0	14.3	17.2	15.8	13.5	10.0	7.5	5.1		
		MCDB	4.8	5.1	7.0	11.3	16.3	18.8	21.5	18.8	15.8	11.5	8.9	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	3.4	4.5	6.4	7.7	7.6	7.1	6.1	4.9	3.8	3.2	3.0		
		MCDBR	4.3	4.3	5.8	8.5	10.8	11.1	10.8	8.6	6.8	5.0	4.4	4.3		
		MCWBR	3.4	3.1	3.6	4.4	5.6	5.5	4.7	4.2	3.5	2.9	3.1	3.7		
		MCDBR	4.2	4.2	5.1	7.9	10.3	9.8	9.8	7.7	6.0	4.1	4.1	4.2		
	5% WB	MCWBR	3.7	3.4	3.6	4.4	5.6	5.3	4.5	4.1	3.4	3.2	3.6	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.300	0.335	0.372	0.374	0.368	0.382	0.376	0.348	0.325	0.279	0.239		
	taud		2.091	2.173	2.154	2.133	2.303	2.383	2.373	2.386	2.454	2.400	2.189	2.018		
	Ebn at Noon		482	681	776	815	850	862	837	813	775	663	463	342		
	Edh at Noon		43	72	102	126	115	108	107	98	77	59	39	29		
(44) All-Sky Solar Radiation	RadAvg		0.26	0.79	2.02	3.82	4.81	4.91	4.59	3.57	2.25	1.06	0.35	0.17		
	RadStd		0.06	0.20	0.38	0.61	0.75	0.59	0.55	0.33	0.26	0.16	0.06	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page